

Age-related patterns of resting EEG power in infancy: Associations with prenatal socioeconomic disadvantage

Aislinn Sandre ^a, Sonya V. Troller-Renfree ^b, Melina Amarante ^a, Amanda M. Dettmer ^c,
Jerrold S. Meyer ^d, Kimberly G. Noble ^{a,*}

^a Department of Biobehavioral Sciences, Teachers College, Columbia University, New York, NY 10027, USA

^b Department of Human Development, Teachers College, Columbia University, New York, NY 10027, USA

^c Yale Child Study Center, Yale School of Medicine, New Haven, CT 06519, USA

^d Department of Psychological and Brain Sciences, University of Massachusetts, Amherst, MA 01003, USA

ARTICLE INFO

Keywords:

Socioeconomic status
Brain development
Electroencephalography (EEG)
Resting brain activity
Infancy

ABSTRACT

The brain develops rapidly during the prenatal period and first two years of life, making it particularly sensitive to environmental influences. Family socioeconomic disadvantage is one environmental factor that may shape the development of brain function in infancy. However, it is unclear how brain function changes across infancy or whether prenatal family socioeconomic disadvantage is associated with age-related differences in brain function during this period. Here, we examine whether resting electroencephalography (EEG) power (theta, alpha, beta, and gamma) shows linear and/or non-linear age-related patterns across four assessments from 1 to 18 months of age ($N = 165$), and whether these patterns are moderated by prenatal family socioeconomic disadvantage. We find that lower-frequency (relative theta) and higher-frequency (relative alpha, beta, and gamma) power show non-linear age-related patterns during the first 18 months of life. Prenatal family socioeconomic disadvantage moderates these patterns, such that infants from lower-income families show less steep age-related decreases in lower-frequency (relative theta) power and less steep increases in higher-frequency (relative beta) power. These associations hold when adjusting for other prenatal and postnatal experiences, as well as infant demographic and health-related factors. These data suggest that lower prenatal family income is associated with age-related differences in brain function during infancy.

1. Introduction

The brain develops rapidly during the prenatal period and first two years of life, undergoing substantial structural and functional changes that lay the foundation for subsequent brain development (Gillmore et al., 2018). The tremendous plasticity of the developing brain during this time makes it especially sensitive to environmental influences, with lasting effects on children's cognitive, socio-emotional, and language development, as well as their lifelong health (Dawson et al., 2000; Monk et al., 2019; Van den Bergh et al., 2020). Family socioeconomic status (SES)—commonly defined as parental educational attainment, occupational prestige, and/or family income—during the prenatal period is one environmental factor that may shape the development of early brain function (Nielsen et al., 2024), likely through a variety of experiences (e. g., maternal stress, toxin exposure, nutrition, and access to prenatal care;

Hackman et al., 2010; Herzberg and Smyser, 2024). However, it is unclear how brain function changes across infancy or whether prenatal family SES is associated with age-related differences in brain function during this period. A better understanding of normative age-related patterns in early brain function, and how prenatal SES influences these patterns, is essential for informing preventive interventions and policies that support infants' long-term development and health.

Electroencephalography (EEG) is a useful tool for understanding age-related patterns in early brain function (Anderson and Perone, 2018). EEG provides a direct, non-invasive measure of electrical brain activity, and is commonly recorded at rest in infants. At rest, EEG oscillations reflect network-level rhythms that coordinate neural activity across brain regions, supporting cognitive, social, and affective processes (Cuevas and Bell, 2022). Resting EEG can be characterized along two key dimensions: frequency and power. "Frequency" refers to oscillatory

* Correspondence to: Department of Biobehavioral Sciences, Teachers College, Columbia University, 525W 120th Street, Russell Hall 21, New York, NY 10027, USA

E-mail address: kgn2106@tc.columbia.edu (K.G. Noble).

<https://doi.org/10.1016/j.dcn.2025.101611>

Received 18 February 2025; Received in revised form 2 September 2025; Accepted 6 September 2025

Available online 8 September 2025

1878-9293/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

brain activity that occurs at different rates across the brain and is categorized into different frequency bands (Saby and Marshall, 2012). Some bands correspond to slower oscillations (e.g., theta), while others reflect faster oscillations (e.g., alpha, beta, and gamma). “Power” refers to the amount of activity within a given frequency band, with higher power indicating greater synchronized electrical activity generated by the brain (Saby and Marshall, 2012; Tan et al., 2024).

Studies examining resting EEG in early childhood typically quantify power using two metrics: absolute power and relative power. Absolute power refers to the raw amount of electrical activity within a specific frequency band, while relative power represents the proportion of power in a specific frequency band relative to the total power across all frequency bands. Both measures have advantages and limitations. For instance, absolute power is highly sensitive to non-neural anatomical factors such as skull thickness, scalp impedance, and hair density (Benninger et al., 1984; Cuffin, 1993; Hagemann et al., 2008), which can introduce noise and may obscure developmental differences. In contrast, relative power normalizes across the EEG spectrum, making it less sensitive to these anatomical factors (Clarke et al., 2001; Marshall et al., 2004). However, because relative power is calculated as a proportion, changes in one frequency band can influence estimates in others, complicating the interpretation of developmental changes. Despite these limitations, we focus on relative power as it may more robustly capture developmental differences by minimizing inter-individual variability.

Because variation in lower- and higher-frequency brain activity is linked to children’s cognitive (Brito et al., 2016; Marshall et al., 2008; Williams et al., 2012), language (Gou et al., 2011) and socio-emotional development (Brito et al., 2019), a growing body of research has sought to chart normative age-related patterns in resting brain activity across early childhood to better understand how it supports these skills. This work suggests that, among typically developing children, relative power in lower-frequency bands (i.e., theta) tends to decrease, while relative power in higher-frequency bands (i.e., alpha, beta, gamma) tends to increase from infancy to middle childhood (Marshall et al., 2002; Perone et al., 2018; Takano and Ogawa, 1998). However, much of this research has relied on cross-sectional designs (Anderson and Perone, 2018; Perone et al., 2018; Takano and Ogawa, 1998) or has assumed linear age-related trends in brain activity (Hofstee et al., 2024; Marshall et al., 2002). Yet, early brain development is a non-linear process (Chen et al., 2021; Liu et al., 2023), necessitating longitudinal investigations that consider non-linear age-related variation.

To date, there is limited longitudinal research examining whether lower- and higher-frequency EEG power show non-linear age-related patterns during the first two years of life—and most existing work has focused on absolute power (e.g., Tierney et al., 2012; Wilkinson et al., 2023). Emerging evidence suggests that relative power may also change non-linearly across infancy. For example, Rico-Picó et al. (2023) found that lower-frequency (relative theta) power rapidly decreased from 6 to 9 months, followed by a slower decline or plateau from 9 to 16 months. In contrast, higher-frequency (relative alpha and beta) power rapidly increased from 6 to 9 months, then showed a slower increase or plateau through 16 months. This suggests that lower- and higher-frequency relative power undergo rapid changes during the first year of life, followed by slower and more stable decreases or increases, respectively, in the second year. Nonetheless, further longitudinal research—especially focused on relative power—is needed to determine whether these patterns follow linear or non-linear trends across the first two years of life, as this may help identify early biomarkers of children’s future developmental outcomes.

Beyond evidence of developmental change, emerging evidence suggests that family SES is associated with differences in resting brain activity during infancy. For instance, infants exposed to socioeconomic disadvantage tend to show more lower-frequency (theta) power and less higher-frequency (alpha, beta, and gamma) power, compared to infants from more advantaged circumstances (Brito et al., 2020; Cantiani et al., 2019; Otero, 1997; Otero et al., 2003; Pierce et al., 2019; Sandre et al.,

2024; Tomalski et al., 2013). Such patterns of brain activity emerge as early as one month of age (Sandre et al., 2024) and persist through childhood and early adolescence (Harmony et al., 1988; Maguire and Schneider, 2019; Otero, 1997). Moreover, boosting family income through unconditional cash transfers to families residing in poverty may cause increased higher-frequency power in infants (Troller-Renfree et al., 2022). However, much of this research has relied on cross-sectional designs and focused primarily on postnatal disadvantage (Brito et al., 2020; Otero, 1997; Pierce et al., 2019; Tomalski et al., 2013). Thus, it remains unclear whether prenatal family SES is associated with differences in age-related patterns of brain activity during infancy, which is crucial for understanding when and how strongly these differences emerge and persist across early development.

Additionally, family SES is a multi-dimensional construct encompassing both correlated and independent economic (e.g., family income) and social (e.g., parental education) factors (Duncan and Magnuson, 2012), each of which may show distinct associations with age-related patterns of brain activity during infancy. Consistent with this, a recent longitudinal study found that lower family income—but not maternal education—was associated with less absolute power in both lower-frequency (theta) and higher-frequency (alpha, beta, gamma) power at 2–3 months, as well as steeper increases in lower-frequency (theta) and higher-frequency (alpha) power from 3 to 9 months (Wilkinson et al., 2023). These findings suggest that income may have particularly strong effects on early brain development. However, this work focused on postnatal SES and absolute EEG power only. Given that few longitudinal studies have simultaneously considered both prenatal family income and parental education, there is a need to examine whether these SES factors show similar or distinct associations with age-related patterns of brain activity during the first two years of life.

In this prospective longitudinal study, we aimed to 1) examine whether resting brain activity shows linear or non-linear age-related patterns during the first 18 months of life, and 2) assess whether prenatal family socioeconomic factors—specifically income-to-needs and parental education—moderate these patterns. We hypothesized that resting brain activity would show non-linear age-related changes, characterized by rapid decreases in lower-frequency (relative theta) power and rapid increases in higher-frequency (relative alpha, beta, and gamma) power during the first year of life, followed by slower decreases and increases in the second year, respectively. Additionally, we predicted that prenatal family SES would moderate associations between age and resting brain activity, such that infants from families with higher income-to-needs ratios and more years of parental education would show more rapid decreases in lower-frequency (relative theta) power and more rapid increases in higher-frequency (relative alpha, beta, and gamma) power across the first 18 months of life compared to infants from more disadvantaged families. Exploratory analyses were conducted to examine whether associations between prenatal family SES and patterns of resting brain activity were stronger over certain regions of the scalp (i.e., frontal, central, parietal, temporal, occipital). Finally, we tested whether any significant associations between prenatal family SES and age-related changes in brain activity held when adjusting for various prenatal and postnatal experiences, infant demographics, and health-related factors that tend to co-vary with SES (Hackman et al., 2010).

2. Methods

2.1. Participants

Participants were 209 mother-infant dyads from the New York City metropolitan area that were part of a larger longitudinal study examining associations between early experiences and child development (Sandre et al., 2024; Sperber et al., 2023; Troller-Renfree et al., 2025). Mothers were recruited during their third trimester of pregnancy from local prenatal clinics, community events, and social media, and were

intentionally recruited to span a range of educational attainment, from having less than a high school education to holding an advanced degree. Recruitment occurred over two time periods because of a temporary interruption in data collection due to the COVID-19 pandemic: the first cohort of mothers (Cohort 1) was recruited from June 2019 through March 2020 ($N = 93$) and the second cohort of mothers (Cohort 2) was recruited from August 2021 to September 2022 ($N = 116$). Inclusion criteria for mothers included: 18 years of age or older, at least 35 weeks pregnant, carrying a singleton fetus with no known neurological or developmental issues, and ability to speak either English or Spanish. Eligible mothers were invited to participate in a prenatal visit in our lab or their home.

After birth, mother-infant dyads were invited to participate in follow-up home or lab visits at four time points: approximately 1, 6, and 12 months of age, as well as at either 15 or 18 months of age (families were invited to complete an 18-month visit only if they did not participate in the 15-month visit). Inclusion criteria for infants included a gestational age of at least 37 weeks and no known neurological or developmental issues at birth. Families in Cohort 2 were invited to all four time points to complete EEG collection. In contrast, families in Cohort 1 were invited to complete EEG collection only at the 1-month visit; all subsequent visits were conducted remotely due to the pandemic. Across all time points, EEG data from eight infants were excluded from analyses due to either poor-quality recordings or an insufficient number of artifact-free epochs (see details below). Additionally, one infant was excluded due to a technical error during EEG recording (i.e., a corrupted file). Of the recruited sample of 209 mother-infant dyads, usable EEG data was collected from 160 infants at the 1-month assessment, 81 infants at the 6-month assessment, 80 infants at the 12-month assessment, and 65 infants at the 15- or 18-month assessments. The present study included infants for whom we had usable EEG data for at least one time point. Therefore, the final sample consisted of 386 resting EEG recordings collected from 165 infants (64 from Cohort 1 and 101 from Cohort 2; see [Supplementary Fig. S1](#)). Descriptive statistics of sample characteristics are presented in [Table 1](#).

A flow chart of the sample is presented in [Supplementary Fig. S2](#). The Supplementary Results also present the results of a series of independent t-tests and chi-square tests, which examined whether study variables differed between families who had EEG data collected for at least one time point ($N = 165$) and those who did not ($N = 44$), as well as between families recruited before (Cohort 1, $N = 64$) and after the pandemic (Cohort 2, $N = 101$) among those who provided EEG data at least once ([Supplementary Table S1](#)). All mothers provided written informed consent for their family's participation in the study. Research procedures were approved by the Institutional Review Board of Teachers College, Columbia University.

2.2. Measures

[Supplementary Fig. S3](#) presents an overview of the measures collected at the prenatal, 1-month, 6-month, 12-month, and 15/18-month time points. [Supplementary Table S2](#) presents descriptive statistics of study variables. For additional details about study measures, see the Supplementary Results.

2.2.1. Prenatal family socioeconomic status

We measured two indices of SES prenatally via questionnaire: average parental education and family income-to-needs (ITN) ratio. Average parental education was calculated by averaging the number of years of education attained by the mother and the second parent. Response options for the parental education measure ranged from 0 to 22 years, with anchors at specific milestones (e.g., 0 = no formal schooling, 12 = high school graduate/GED, 14 = associates degree or trade school, 16 = bachelor's degree, 18 = master's degree, 22+ = PhD, MD, JD, or other professional degrees). ITN was calculated by dividing total household income by the U.S. poverty threshold for the respective

Table 1

Descriptive statistics of sample characteristics.

Variable	Mean (Standard Deviation; Range)	N (%)
Mother age at birth (years)	32.35 (5.58; 19 – 45)	-
Infant gestational age at birth (weeks)	39.46 (1.08; 37.00 – 42.86)	-
Infant weight at birth (lbs)	7.39 (1.03; 5.00 – 9.94)	-
Infant age at 1-month EEG (months)	1.30 (0.49; 0.49 – 2.96)	-
Infant age at 6-month EEG (months)	6.77 (0.85; 5.66 – 9.44)	-
Infant age at 12-month EEG (months)	12.61 (0.87; 11.57 – 16.27)	-
Infant age at 15/18-month EEG (months)	16.54 (1.28; 15.06 – 20.35)	-
Infant sex		
Male	-	75 (45)
Female	-	90 (55)
Infant race		
White	-	65 (40)
Black or African American	-	41 (25)
Asian	-	4 (2)
American Indian/Alaska Native	-	2 (1)
Other	-	41 (25)
Refused	-	11 (7)
Infant ethnicity		
Hispanic or Latino	-	74 (45)
Not Hispanic or Latino	-	86 (52)
Refused	-	4 (3)
Prenatal family income (USD)	172,920.74 (325,010.99; 0.00–2,563,500.00)	-
Prenatal family income-to-needs	7.06 (9.44; 0.00 – 43.74)	-
Prenatal parental education (years)	14.97 (3.23; 6 – 22)	-
Usable EEGs		
1-month	-	160 (97)
6-months	-	81 (49)
12-months	-	80 (48)
15/18-months	-	65 (39)
Total Usable EEGs Collected	-	386 (58)
Usable EEGs per infant	2.34 (1.34; 1 – 4)	-

family size for the year of data collection. An ITN below 1 indicates that a family is living below the federal poverty threshold, whereas an ITN above 1 indicates that a family is living above the federal poverty threshold. Three mothers had outlying (>3 SDs) ITN values, and so these values were winsorized prior to analyses to preserve power for analyses. The ITN for the sample ranged from 0 to 43.74, with a median ITN value of 2.81 (IQR=9.28). Twenty-seven percent ($N = 44$) of the sample reported an ITN below the poverty line, and eight percent ($N = 14$) of the sample reported ITN values between 1 and 2, considered to be within the “near-poor” range. Additionally, ten mothers reported a family income of zero dollars. ITN values were positively skewed, and so ITN values were natural log-transformed ($\ln$) for all analyses. To enable log transformation, one dollar was added to all income values prior to calculating ITN. [Supplementary Fig. S4](#) presents histograms of family income and ITN, and [Supplementary Fig. S5](#) presents a histogram of parental education. Family ITN was also measured at 12-months. [Supplemental Fig. S.6](#) presents a spaghetti plot depicting changes in family ITN from the prenatal to 12-month time points.

2.2.2. Maternal perceived stress

Maternal perceived stress was assessed using the Perceived Stress Scale (PSS-10; [Cohen et al., 1983](#)) at the prenatal, 6-month, 12-month, and 15/18-month time points, and the Life Events Scale (LES; [Sarason et al., 1978](#)) at the prenatal and 12-month time points. Consistent with prior research from our lab ([Merz et al., 2019](#); [Simon et al., 2021](#)), we created a “Prenatal Perceived Stress Composite” and a “Postnatal Perceived Stress Composite” by conducting separate exploratory factor

analyses for the prenatal and postnatal stress measures, using principal factor extraction and Varimax rotation. Higher scores on the Prenatal and Postnatal Composites indicate greater maternal perceived stress.

2.2.3. Maternal physiological stress

Maternal physiological stress was assessed prenatally via hair cortisol. A small hair sample was collected from a standardized area of the posterior vertex of the mother's scalp, with each sample weighing at least 15 mg. Each hair sample was trimmed to be approximately 3 cm long (measured from the end closest to the root), thereby containing cortisol deposited during roughly the past three months (based on an estimate of 1 cm hair growth per month; Greff et al., 2019). Three mothers were excluded from analyses because they were currently using steroid medications. Additionally, two mothers had outlying (>3 SDs) hair cortisol values, and so these values were winsorized prior to analyses. Hair cortisol values were natural log-transformed ($\ln$) to correct for skew. There were no significant associations between hair cortisol and hair washing frequency, use of oral contraceptives, or use of hair dye ($ps > .65$).

2.2.4. Maternal vocabulary skills

Maternal vocabulary skills were measured prenatally using the Picture Vocabulary Test (PVT) of the NIH Toolbox Cognition Battery (Gershon et al., 2014). The PVT assesses receptive vocabulary via auditory comprehension of single words that are graded in difficulty using an auditory word-picture matching paradigm. Mothers' performance on the PVT was converted to a theta score (ranging from 4 to -4), based on item response theory.

2.2.5. Parent-child activities

Maternal engagement in infant learning activities was assessed at 6 months using the Parent-Child Activities Index (Rodriguez and Tamis-LeMonda, 2011). Mothers were asked about the frequency of which they (and the second parent) engaged in reading, telling stories, building things, and participating in playgroups with their infants. Scores for maternal and second parent activities were summed separately and averaged to compute an overall parent-child activities score. Higher scores on the parent-child activities index indicate more engagement in learning activities.

2.2.6. Home language input

The home language environment was assessed at 6-months using the Language Environment Analysis (LENA) system, an automated device that can record a child's language environment for up to 16 h. We used LENA's validated speech-identification algorithms to segment the audio and estimate the number of adult words spoken to the child. Mean hourly adult word counts were calculated by dividing the adult word count by the total recording duration (in hours) for each participant.

2.2.7. Feeding style

At one-, six-, and twelve-months, mothers were asked how they were currently feeding their infant (e.g., "What do you currently feed your baby?"). At one- and six-months, response options included "Only breast milk", "Only baby formula", and "Both breast milk and baby formula". At twelve months, an additional option was provided: "Other milks (e.g., cow's milk)".

2.2.8. Nutrition

At 12 months, mothers reported their child's daily intake of fruits, vegetables, sweets (e.g., sweetened cereals, fruit bars), and sweetened drinks (e.g., juice, chocolate milk) using an adapted version of the Los Angeles County WIC Survey (Davis Research LLC, 2017). Frequency was assessed via a Likert scale ranging from 0 ('never') to 5 ('5 times per day'). Responses were summed to create two daily food intake indices: a 'healthy foods' index (fruits and vegetables) and an 'unhealthy foods' index (sweets and sweetened drinks), representing total daily frequency

for each food type.

2.2.9. Infant demographic and health-related factors

Mothers reported infants' gestational age at birth, birth weight, sex, race, ethnicity, and age at the 1-month visit. Infant ages at the 6-, 12-, and 15/18-month visits were also collected via questionnaires.

2.3. Electroencephalography data acquisition and processing

Resting EEG was recorded at 1, 6, 12, and either 15 or 18 months using a 128-channel HydroCel Geodesic Sensor Net (Magstim Electrical Geodesic Inc., Eugene, OR, USA) with a Net Amps 400 amplifier. Electrode impedances were kept below 50 k Ω whenever possible. Data were sampled at 1000 Hz and referenced online to Cz. The EEG collection procedure is detailed in the Supplementary Results. EEG data were analyzed offline using the EEGLAB toolbox (Delorme and Makeig, 2004; Gershon 2022.1), MATLAB (Version R2021b; The MathWorks, Natick, MA), and the MADE pipeline (Debnath et al., 2020). The outermost ring of electrodes (i.e., electrodes located near the base of the skull) were removed since they are heavily susceptible to eye, face and head movements. Data were then high-pass filtered at 0.30 Hz and low-pass filtered at 50 Hz. Bad channels were identified and removed using the EEGLAB plug-in FASTER (Nolan et al., 2010). Infants with globally bad channels across all epochs were excluded from further processing (removed $N = 3$). Ocular artifacts and generic noise were removed using the standard MADE ICA removal procedure (Debnath et al., 2020). Artifactual ICs were removed from the original dataset by using the Adjusted-ADJUST algorithm (Leach et al., 2020).

Subsequently, data were epoched in one-second segments with 50 % (500 ms) overlap. Epochs where any of the frontal electrodes exceeded a voltage threshold of ± 150 μ V were rejected to ensure the removal of ocular artifacts. For the remaining epochs, bad channels were identified if the electrode exceeded the voltage threshold, and epochs were rejected when more than 20 % of channels were interpolated. Infants with artifactual epochs across all segments (i.e., more than 20 % of channels interpolated on every epoch) were excluded from further processing (removed $N = 4$). All rejected channels were interpolated. Data were then re-referenced to an average reference.

Following pre-processing, a Fast Fourier Transform with a Hanning Window was applied to the epoched data. Power spectral density was first calculated in linear units (μ V²) for each electrode and epoch. Power spectra were then averaged across usable epochs for each electrode. Subsequently, a $\log_{10}$ transformation (specifically, $\log_{10}(1 + \mu$ V²) was applied to the averaged power spectra to reduce skew and improve distributional properties (the $+1$ ensures all values remain positive). Absolute power for each frequency band—theta (3–5 Hz), alpha (6–9 Hz), beta (13–19 Hz), and gamma (21–45 Hz)—was computed by averaging the log-transformed power values within the corresponding frequency ranges. These bands were selected to align with prior infant EEG research (Rico-Picó et al., 2023; Sandre et al., 2024; Tomalski et al., 2013). Although there is some variability in the exact frequency ranges used to define gamma power across studies—for example, some have labeled 20–30 Hz as high beta (Arnett et al., 2024; Wilkinson et al., 2024)—we selected this range for gamma power to align with prior studies in infants of a similar age to our sample (Brito et al., 2020; Jensen et al., 2021; Tomalski et al., 2013; Troller-Renfree et al., 2023).

Relative power for each frequency band was computed by dividing the band-averaged power by the sum of the band-averaged powers across all four bands. For example, relative alpha power was calculated as: Relative alpha = $\text{mean}(\log_{10}(1 + \text{AlphaFreqs})) / (\text{mean}(\log_{10}(1 + \text{ThetaFreqs})) + \text{mean}(\log_{10}(1 + \text{AlphaFreqs})) + \text{mean}(\log_{10}(1 + \text{BetaFreqs})) + \text{mean}(\log_{10}(1 + \text{GammaFreqs})))$. Absolute and relative power values were first computed per electrode and then averaged across all electrodes to obtain whole-brain estimates. We also calculated both absolute and relative EEG power for each frequency band within five regions of interest (ROIs): frontal, central, parietal, temporal, and

occipital (see Sandre et al., 2024 for electrodes included in our regional scorings). Absolute power for each region was computed by averaging the power across all electrodes within that ROI for each frequency band. To calculate relative power, we then divided the absolute power in each frequency band by the total power summed across all frequency bands within the same ROI. This process was repeated for each of the five ROIs. Given that relative power may be a more sensitive measure of developmental change in EEG power as it minimizes individual differences in skull thickness, resistance and impedances that impact absolute power (Clarke et al., 2001; Marshall et al., 2004), our primary analyses focused on relative power.

To determine the minimum number of epochs needed for reliable power estimates at each time point, we used the Spearman-Brown split-half correlation procedure with multiple iterations (Leach et al., 2020; Troller-Renfree et al., 2021). The results of these analyses are reported in Supplementary Table S3. One infant had too few artifact-free epochs at 12-months to provide reliable EEG power estimates and was therefore excluded from our analyses. Descriptive statistics of the number of artifact-free epochs at each EEG assessment are reported in Supplementary Table S4.

2.4. Missing data

Supplementary Table S2 presents the number and percentage of mother–infant dyads with available data for each study measure relative to the final sample ($N = 165$). To assess whether missing data in our independent variables and covariates were missing completely at random (MCAR), we conducted Little's MCAR test (Little, 1988), which indicated that the data were indeed MCAR, $\chi^2(1689) = 1567.30$, $p > .05$. To address this missingness, we conducted multilevel imputation for independent variables and covariates; EEG data were not imputed. Additional details on this procedure are provided in the Supplementary Results.

2.5. Statistical analyses

All statistical analyses were conducted in SPSS (Version 29) or R (Version 4.4.1). To examine age-related patterns in infant resting brain activity, we used Generalized Additive Mixed-Effects Models (GAMMs; Wood, 2004), which flexibly model both linear and nonlinear relationships by fitting smooth functions to predictors. Smooth functions are constructed from a linear combination of weighted basis functions and are penalized for increasing complexity (Wood, 2004). Instead of beta coefficients, GAMMs compute effective degrees of freedom (EDF) for smooth terms, which specify the complexity of the relationship between predictor and outcome variables. An EDF of 1 is linear and higher EDFs indicate more nonlinearity.

All GAMMs were structured so that EEG power, infant age, and the number of artifact-free epochs at each time point were nested within participants. In our first set of models, we examined age-related differences in resting brain activity by fitting separate GAMMs for whole-brain relative theta, alpha, beta, and gamma power as the dependent variable. Infant age (in months) was included as a smooth term, with cohort number (0 = Cohort 1, 1 = Cohort 2), and number of artifact-free epochs as linear covariates. To identify periods of significant change in associations between age and whole-brain relative power, we calculated the first derivative of the smooth age term using the *gratia* package (Simpson, 2024). Intervals of significant age-related change were identified when the simultaneous confidence interval of the derivative did not include zero. Following this, we conducted follow-up regional analyses by fitting separate GAMMs for each of our five whole-brain ROIs, with each ROI entered as the dependent variable. We defined our GAMMs as follows:

Whole-brain: Relative Power $\sim s(\text{Age}, k = 4) + \text{Covariates}$

Regional: Relative Power ROI $\sim s(\text{Age}, k = 4) + \text{Covariates}$

In our second set of models, we examined whether prenatal family

SES moderated age-related differences in resting brain activity by fitting separate GAMMs for whole-brain relative theta, alpha, beta, and gamma power as the dependent variables. Infant age and prenatal SES (family ITN or parental education) were included as smooth terms, along with a tensor product interaction term for age and prenatal SES. This interaction term allows the smooth term for age to vary as a fully non-linear function of prenatal SES. Cohort number and the number of artifact-free epochs were included as linear covariates. Because nonlinear moderations preclude follow-up with simple slopes (Wood, 2017), significant interactions were probed by examining associations between prenatal SES and age-related differences in whole-brain relative power separately in infants from families with higher and lower SES, as defined by a median split. To determine whether associations between family SES and age-related differences in infant brain activity were stronger over certain regions of the scalp (i.e., frontal, central, parietal, temporal, and occipital), we conducted separate GAMMs for each of our five ROIs, using the same predictors and covariates as in our whole-brain models. We defined our GAMMs as follows:

Whole-brain: Relative Power $\sim s(\text{Age}, k = 4) + s(\text{Prenatal Family SES}, k = 4) + \text{ti}(\text{Age}, \text{Prenatal Family SES}, k = 4) + \text{Covariates}$

Regional: Relative Power ROI $\sim s(\text{Age}, k = 4) + s(\text{Prenatal Family SES}, k = 4) + \text{ti}(\text{Age}, \text{Prenatal Family SES}, k = 4) + \text{Covariates}$

All GAMMs included a random intercept for participant with age splines restricted to four basis functions ($K=4$) for flexibility. Given that GAMMs can model both linear and non-linear associations, in instances where there is no non-linear association between age and resting brain activity, the smooth term for age was penalized down to a linear term. Models used thin plate regression splines for smooth terms, with smoothing parameters estimated via restricted maximum likelihood (REML). GAMMs were computed using the *mgcv* package.

To account for multiple comparisons, P -values from all related GAMMs were corrected using the false discovery rate (FDR) method with the *p.adjust* package. For whole-brain analyses, FDR corrections were applied across the four frequency bands (relative theta, alpha, beta, and gamma), resulting in 4 corrections for whole-brain relative power. For regional analyses, FDR corrections were applied across 5 ROIs—frontal, central, parietal, temporal, and occipital—within each frequency band, resulting in 5 comparisons per frequency band. This yielded a total of 20 comparisons across all regions and frequency bands. The same FDR correction approach was applied to whole-brain and regional absolute power.

2.6. Sensitivity analyses

Given that family SES often covaries with various early experiences and family characteristics (Bradley and Corwyn, 2002; Farah, 2017; Hackman et al., 2010; Hoff and Laursen, 2019; Noble et al., 2021; Ursache et al., 2017), we tested the robustness of associations between prenatal SES and age-related patterns in infant resting brain activity by re-running our whole-brain relative power models with a set of covariates that have established links to SES (Hackman et al., 2010). Specifically, we included maternal prenatal perceived stress and physiological stress (i.e., hair cortisol), receptive vocabulary, maternal postnatal perceived stress, postpartum exclusive breastfeeding at one, six and 12-months (1 = exclusively breastmilk fed, 0 = exclusively formula fed, 0 = mixed breastmilk and formula fed), exclusive formula feeding at one, six and 12-months (1 = exclusively formula fed, 0 = exclusively breastmilk fed, 0 = mixed breastmilk and formula fed), exclusive other milk fed at 12months (1 = exclusively other milk fed, 0 = exclusively breastmilk fed, 0 = exclusively formula fed, 0 = mixed breast milk, formula or other-milk fed), parent-child activities, adult word count, healthy and unhealthy foods consumed per day, postnatal ITN, infant sex (0 = male, 1 = female), race (0 = White, 1 = Non-white), ethnicity (0 = Non-Hispanic/Latino, 1 = Hispanic/Latino), weight at birth (in lbs), and gestational age at birth (in weeks). All covariates were entered simultaneously into each model. We modeled all covariates as

linear terms to prioritize parsimony and minimize the risk of model overfitting, given the absence of strong a priori hypotheses about the functional form of these associations.

Smooth terms for both prenatal family ITN and parental education were included in our models to examine their unique associations with infant whole-brain relative power. Although prenatal and postnatal family ITN were significantly correlated ($r = .65, p < .01$), as were prenatal ITN and parent education ($r = .53, p < .01$) they did not violate assumptions of multicollinearity ($VIF < 7$; Kim, 2019).

3. Results

3.1. Age-related patterns of infant whole-brain relative power

We first examined whether infant whole-brain relative power showed linear or non-linear age-related patterns during the first 18 months after birth. Results indicated that whole-brain relative theta ($F_s(\text{Age}) = 215.30$, $\text{EDF} = 2.84$, $R^2_{\text{partial}} = 0.70$, $p_{\text{FDR}} < .001$), alpha ($F_s(\text{Age}) = 215.20$, $\text{EDF} = 2.77$, $R^2_{\text{partial}} = 0.70$, $p_{\text{FDR}} < .001$), beta ($F_s(\text{Age}) = 35.05$, $\text{EDF} = 2.80$, $R^2_{\text{partial}} = 0.29$, $p_{\text{FDR}} < .001$), and gamma power ($F_s(\text{Age}) = 41.07$, $\text{EDF} = 2.95$, $R^2_{\text{partial}} = 0.35$, $p_{\text{FDR}} < .001$) showed non-linear age-related trends during the first 18-months after birth, after controlling for covariates. Relative theta power significantly decreased over the first 15.0 months, then plateaued with no significant change between 15.1 and 20.4 months of age (Fig. 1a). Relative alpha power increased during the first 15.3 months and subsequently plateaued with no significant change between 15.5 and 20.4 months (Fig. 1b). Relative beta power increased during the first 8.1 months, then plateaued with no significant change between 8.3 and 20.4 months (Fig. 1c). Relative gamma power increased over the first 7.1 months, decreased after 8.5

months, and then plateaued with no significant change between 13.9 and 20.4 months of age (Fig. 1d). Topographic heat maps showing the scalp distribution of relative power for each of the four frequency bands at each time point are presented in Fig. 2. Additional results are provided in the Supplementary Results, including age-related patterns of regional relative power (Table S5, Fig. S7), whole-brain and regional absolute power (Table S6, Figs. S8 & S9), topographic heat maps of absolute power (Fig. S10), and power spectra for each time point (Figs. S11 & S12).

3.2. Associations between prenatal family ITN and age-related patterns of infant whole-brain relative power

We next examined whether prenatal family ITN moderates age-related patterns of infant whole-brain relative power (results for absolute power are presented in Supplementary Table S7 and Fig. S13 & Fig. S14). As indicated in Table 2, higher prenatal family ITN was marginally associated with less relative theta power ($F_{s(\text{ITN})} = 3.52$, $\text{EDF} = 2.03$, $p = 0.03$, $p_{\text{FDR}} = .06$) and significantly associated with more relative beta power ($F_{s(\text{ITN})} = 4.69$, $\text{EDF} = 2.19$, $p_{\text{FDR}} = .03$) in infants during the first 18-months after birth, after controlling for covariates (Supplementary Fig. S15). Moreover, prenatal family ITN significantly moderated age-related patterns of relative theta power ($F_{ti(\text{Age} \times \text{ITN})} = 3.02$, $\text{EDF} = 3.80$, $R^2_{\text{partial}} = 0.06$, $p_{\text{FDR}} < .05$) and relative beta power ($F_{ti(\text{Age} \times \text{ITN})} = 4.06$, $\text{EDF} = 4.02$, $R^2_{\text{partial}} = 0.07$, $p_{\text{FDR}} < .05$) in infants after controlling for covariates (Table 2). Infants from families with higher prenatal ITN showed a qualitatively more rapid decrease in relative theta power compared to infants from families with lower prenatal ITN (Fig. 3a). This difference emerged around 2 months of age, widened over time, and plateaued by 11 months. In contrast, infants

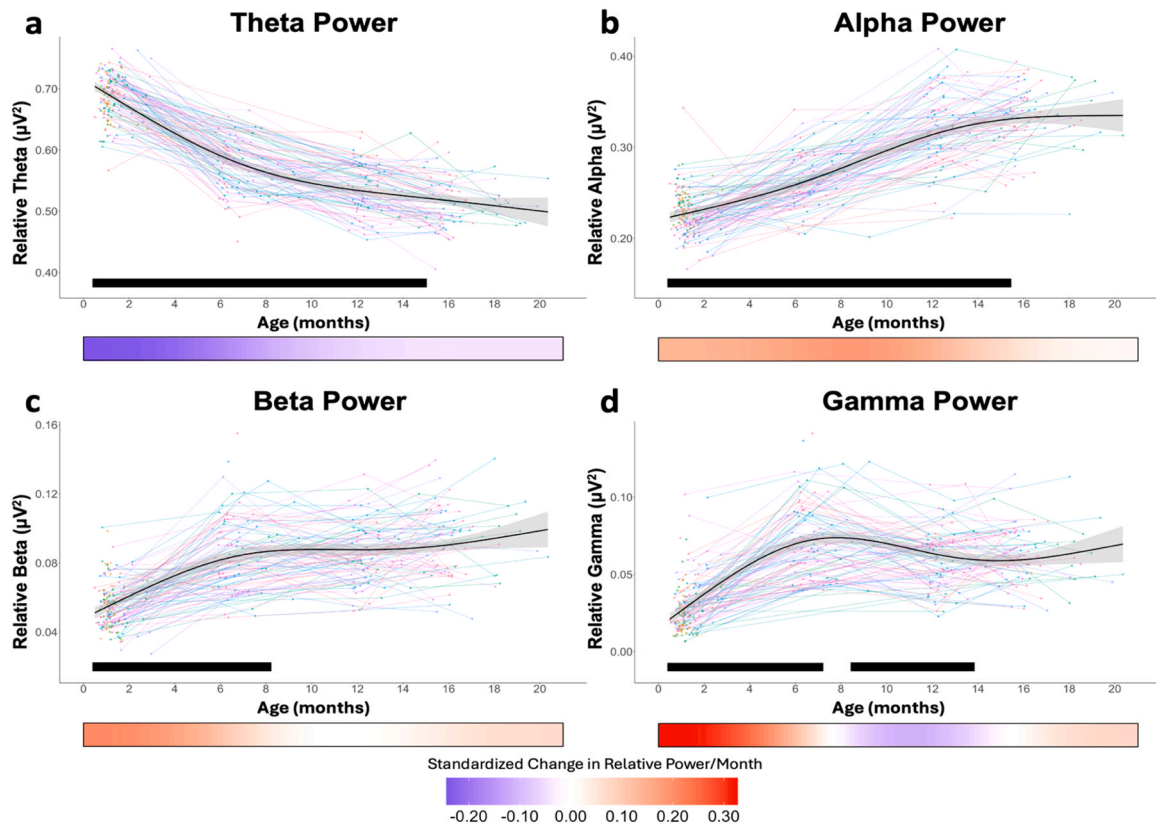


Fig. 1. Age-related patterns of whole-brain relative theta, alpha, beta and gamma power in infants during the first 18 months of life. *Note.* Black lines represent GAMM-predicted age-related patterns of relative power at the group level, with shaded areas indicating 95 % confidence intervals. Colored lines depict individual infants' relative power values across age. Black bars above the x-axes indicate periods of significant change in relative power. Colored bars below the x-axes display the derivative of the fitted age term, with the saturation of the fill representing the standardized value of the derivative. Standardized change in relative power per month is calculated as the change per month divided by the standard deviation across the full age range.

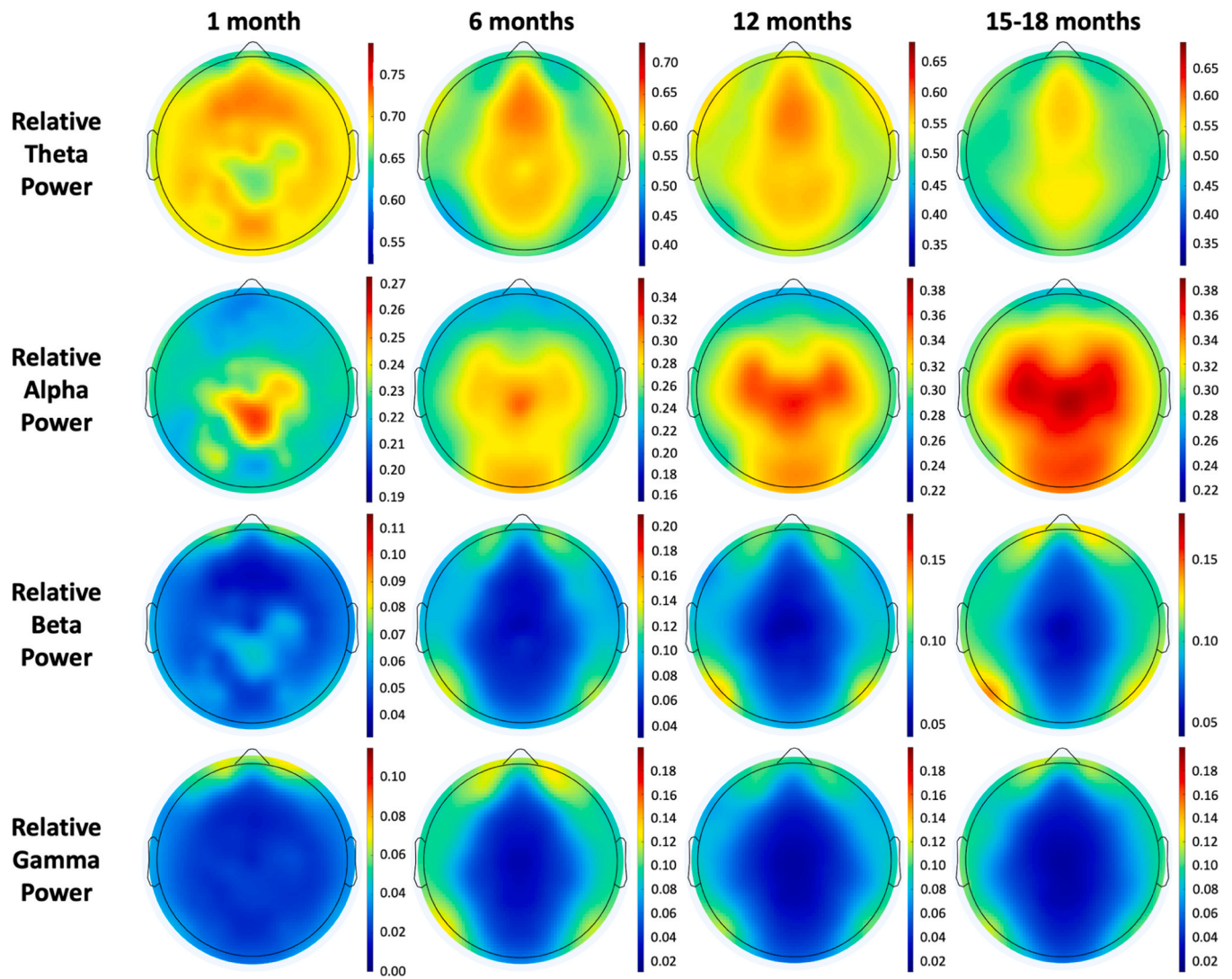


Fig. 2. Topographic heat maps of the distribution of relative theta, alpha, beta, and gamma power across the scalp in infants at each timepoint. Note. Warmer colors represent more power in each respective frequency band.

Table 2

Results of the GAMMs examining associations between prenatal family income-to-needs (ITN), parental education and age-related patterns of whole-brain relative theta, alpha, beta, and gamma power in infants.

N = 165	Relative Theta				Relative Alpha				Relative Beta				Relative Gamma			
Predictors	EDF	F	β	SE	EDF	F	β	SE	EDF	F	β	SE	EDF	F	β	SE
s(Age)	2.83	219.36**	-	-	2.78	211.85**	-	-	2.79	35.41**	-	-	2.95	40.79**	-	-
s(Prenatal family ITN)	2.03	3.52 [†]	-	-	1.00	0.49	-	-	2.19	4.69*	-	-	1.65	0.81	-	-
ti(Age*Prenatal family ITN)	3.80	3.02*	-	-	1.47	0.13	-	-	4.02	4.06*	-	-	2.00	2.61 [†]	-	-
Number of epochs			.10**	.03			.03	.04			-.21**	.05			-.15**	.05
Cohort			.00	.04			-.11*	.05			.05	.06			.15*	.06
	Adjusted R ² = 0.79				Adjusted R ² = 0.66				Adjusted R ² = 0.50				Adjusted R ² = 0.50			
N = 165	Relative Theta				Relative Alpha				Relative Beta				Relative Gamma			
Predictors	EDF	F	β	SE	EDF	F	β	SE	EDF	F	β	SE	EDF	F	β	SE
s(Age)	2.84	221.36**	-	-	2.78	214.74**	-	-	2.78	35.24**	-	-	2.95	40.92**	-	-
s(Prenatal parental education)	1.00	0.69	-	-	1.98	1.63	-	-	1.97	2.46	-	-	1.36	0.91	-	-
ti(Age*Prenatal parental education)	4.22	2.56	-	-	2.72	1.40	-	-	3.21	1.13	-	-	3.21	1.31	-	-
Number of epochs			.09**	.03			.03	.04			-.20**	.05			-.15**	.05
Cohort			.01	.04			-.11*	.05			.05	.06			.15*	.06
	Adjusted R ² = 0.78				Adjusted R ² = 0.67				Adjusted R ² = 0.48				Adjusted R ² = 0.49			

Note. Cohort (0 = Cohort 1, 1 = Cohort 2); ** $p_{FDR} < .01$ * $p_{FDR} < 0.05$ [†] $p_{FDR} < .10$.

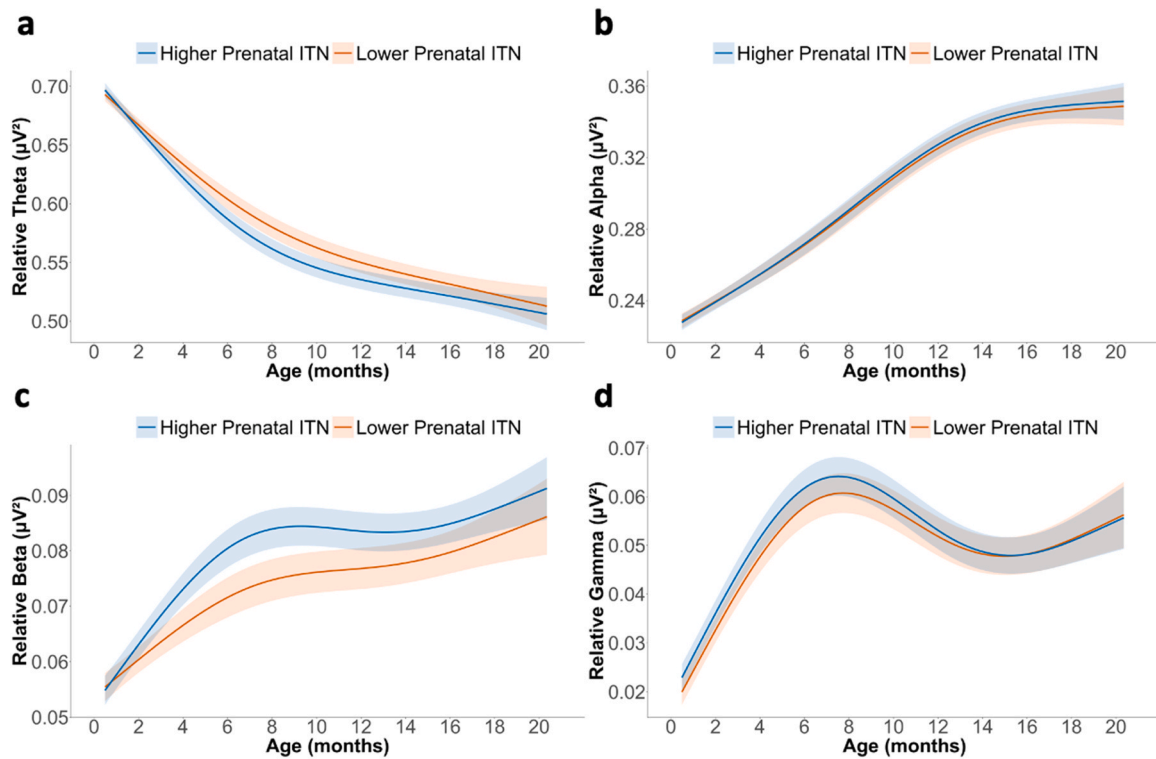


Fig. 3. Associations between prenatal family income-to-needs (ITN) and age-related patterns of relative theta, alpha, beta and gamma power in infants. *Note.* Plots display GAMM-predicted age-related patterns of relative power for infants from families with lower prenatal family ITN ratios (red) and higher prenatal family ITN ratios (blue). Family ITN was modeled continuously, but for visualization purposes, model-predicted values are shown using a median split of ITN. Shaded areas represent 95 % confidence intervals.

from families with higher prenatal ITN showed a qualitatively more rapid increase in relative beta power compared to infants from families with lower prenatal ITN (Fig. 3c). This difference emerged around 1 month of age, widened over time, and plateaued by 12 months. We did not observe any significant main effects of prenatal family ITN, nor interactions between age and prenatal family ITN, on relative alpha or gamma power in infants (Fig. 3b & d).

We repeated our analyses examining associations between prenatal family ITN and age-related patterns of whole-brain relative power, excluding multiply imputed family ITN ratios ($N = 10$; Supplementary Table S8 and Fig. S16) and families with ITN ratios greater than 20 ($N = 12$; Supplementary Table S9 and Fig. S17). The results of these analyses were very similar to those reported here. To evaluate whether our findings were driven by infants who contributed EEG data only at the 1-month timepoint—particularly those overrepresented in Cohort 1—we conducted a sensitivity analysis by repeating our GAMMs for whole-brain relative theta and beta power restricted to Cohort 2 ($N = 101$; Supplementary Table S10 and Fig. S18). This subset reflects the portion of the sample for whom EEG data collection continued post-pandemic and includes a higher proportion of participants with repeated measures. Results from this analysis were consistent with those from the full sample.

3.3. Associations between prenatal family ITN and age-related patterns of infant regional relative power

To better understand whether associations between prenatal family ITN and age-related patterns of relative theta and beta power were stronger over certain regions of the scalp, we conducted follow-up GAMMs for relative power over frontal, central, parietal, temporal and occipital regions (Supplementary Table S11). Results of these analyses indicated that higher prenatal family ITN was associated with a qualitatively more rapid decrease in relative theta power in frontal (F_{ti}

($AgexITN$) = 3.04, $EDF = 4.27$, $R^2_{partial} = 0.07$, $p = .02$, $p_{FDR} = .08$), temporal ($F_{ti(AgexITN)} = 2.38$, $EDF = 4.07$, $R^2_{partial} = 0.05$, $p = .05$, $p_{FDR} = .08$) and occipital regions ($F_{ti(AgexITN)} = 2.78$, $EDF = 2.47$, $R^2_{partial} = 0.04$, $p = .03$, $p_{FDR} = .08$), after controlling for covariates (Supplementary Fig. S19). However, these associations were not significant after FDR correction. In contrast, higher prenatal family ITN was associated with a qualitatively more rapid increase in relative beta power in frontal ($F_{ti(AgexITN)} = 3.68$, $EDF = 4.70$, $R^2_{partial} = 0.09$, $p_{FDR} < .01$), central ($F_{ti(AgexITN)} = 4.68$, $EDF = 3.48$, $R^2_{partial} = 0.08$, $p_{FDR} < .01$), parietal ($F_{ti(AgexITN)} = 4.47$, $EDF = 2.82$, $R^2_{partial} = 0.06$, $p_{FDR} < .01$), temporal ($F_{ti(AgexITN)} = 2.56$, $EDF = 4.19$, $R^2_{partial} = 0.05$, $p_{FDR} < .05$), and occipital regions ($F_{ti(AgexITN)} = 5.33$, $EDF = 1.93$, $R^2_{partial} = 0.03$, $p_{FDR} < .05$) after controlling for covariates (Supplementary Fig. S20).

3.4. No association between prenatal parental education and age-related patterns of infant whole-brain relative power

We next examined whether prenatal parental education moderates age-related patterns of infant whole-brain relative power (results for absolute power are presented in the Supplementary Table S7 and Fig. S21). We did not observe any significant main effects of parental education, nor interactions between age and parental education, in any frequency band in infants after controlling for covariates (Table 2). See Fig. S22 in the Supplementary Results for age-related patterns of infant whole-brain relative power, plotted separately by families' highest level of parental educational attainment.

3.5. Sensitivity analyses

Finally, we conducted sensitivity analyses to test the robustness of our findings by examining whether prenatal family ITN moderates age-related patterns of infant whole-brain relative theta and beta power when adjusting for the influence of other measured prenatal and

postnatal experiences; infant demographic and health-related factors; postnatal family ITN; and parental education (Table 3; results for whole-brain absolute power are presented in Supplementary Table S12 and Fig. S23). We again found that prenatal family ITN significantly moderated age-related patterns of whole-brain relative theta ($F_{ti(Age \times ITN)} = 3.38$, $EDF = 4.17$, $R^2_{partial} = 0.06$, $p_{FDR} < .01$) and relative beta power ($F_{ti(Age \times ITN)} = 4.27$, $EDF = 4.35$, $R^2_{partial} = 0.07$, $p_{FDR} < .01$) in infants after controlling for these covariates. Infants from families with higher prenatal ITN showed a qualitatively more rapid decrease in relative theta power as well as a qualitatively more rapid increase in relative beta power, compared to infants from families with lower prenatal ITN (Supplementary Fig. S24). Moreover, prenatal parental education did not significantly moderate age-related patterns of whole-brain relative theta and beta power when controlling for covariates, indicating that socioeconomic-related differences in infant EEG power were specific to family ITN and not parental education.

4. Discussion

Here we examined whether resting brain activity shows linear or non-linear age-related patterns during the first 18 months of life, and whether prenatal family SES moderates these patterns. Consistent with our hypotheses, both lower-frequency (relative theta) and higher-frequency (relative alpha, beta, and gamma) power showed non-linear age-related changes. Specifically, relative theta power decreased most rapidly during the first 15 months. Relative alpha power increased most rapidly during the first 15 months, while relative beta power increased most rapidly during the first 8 months before plateauing. In contrast, relative gamma power followed a more complex pattern, with a rapid increase during the first 7 months, a decline between 8 and 14 months, and a plateau thereafter. Additionally, prenatal family ITN moderated these non-linear associations between age and resting brain activity,

such that infants from families with higher ITN showed a more rapid decrease in lower-frequency (relative theta) power and a more rapid increase in higher-frequency (relative beta) power over time, compared to infants from families with lower ITN.

These results are consistent with prior research indicating age-related changes in relative EEG power during the first two years of life (Hofstee et al., 2024; Marshall et al., 2002) and further clarify this work by showing that these changes in may be non-linear. EEG power reflects network-level oscillations, with lower-frequency (theta) oscillations supporting long-range communication between networks and higher-frequency (alpha, beta, gamma) oscillations facilitating short-range, localized communication within networks (Von Stein and Sarnthein, 2000). The non-linear patterns of brain activity observed likely reflect multiple dynamic neurodevelopmental processes, which contribute to the integration of more efficient local and large-scale neural networks (Hu et al., 2022; Von Stein and Sarnthein, 2000). For example, increases in relative alpha, beta, and gamma power may reflect increases in myelination, which peaks during the first year and slows thereafter (Dubois et al., 2014), enhancing the speed and synchronization of neuronal signaling. It is important to note, however, that because relative power is a proportional measure—with total power across all frequency bands in the denominator—observed decreases in relative theta power may reflect increases in alpha, beta, and/or gamma power. Indeed, we observed an age-related shift in peak frequencies from the theta to the alpha range, which may also contribute to the decline in theta power (see Supplementary Figs. S11 and S12). This is a key limitation when interpreting age-related changes in relative power and underscores the importance of examining both absolute and relative metrics within the same study. At the regional level, we also observed both linear and non-linear age-related patterns of relative power (see Table S5 and Fig. S7), consistent with evidence that myelination progresses at different rates across brain regions (Dubois et al., 2014;

Table 3

Sensitivity analyses from GAMMs examining associations between prenatal family income-to-needs and age-related patterns of whole-brain relative theta and beta power in infants.

N = 165 Predictors	Relative Theta				Relative Beta			
	EDF	F	β	SE	EDF	F	β	SE
s(Age)	2.83	213.36**	-	-	2.79**	34.29	-	-
s(Prenatal family ITN)	2.28	3.23 [†]	-	-	2.08	2.49 [†]	-	-
s(Prenatal parental education)	1.00	1.39	-	-	1.00	0.00	-	-
ti(Age*Prenatal family ITN)	4.17	3.38*	-	-	4.35	4.27**	-	-
Maternal prenatal perceived stress	-	-	-.01	.04	-	-	.02	.07
Maternal prenatal hair cortisol (ln; pg/mg)	-	-	-.01	.04	-	-	.05	.06
Maternal receptive vocabulary	-	-	.02	.04	-	-	-.07	.07
Infant gestational age at birth (weeks)	-	-	-.05	.03	-	-	.02	.05
Infant weight at birth (lbs)	-	-	.07 [†]	.03	-	-	-.03	.05
Maternal postnatal perceived stress	-	-	.02	.04	-	-	-.04	.07
Parent-child activities	-	-	.03	.03	-	-	-.06	.05
Adult word count (hourly)	-	-	.04	.03	-	-	-.01	.05
Postnatal family ITN	-	-	.06	.04	-	-	.01	.07
Breastfed at 1 month	-	-	-.03	.04	-	-	.04	.06
Formula fed at 1 month	-	-	-.02	.04	-	-	.10	.06
Breastfed at 6 months	-	-	.06	.05	-	-	-.14	.09
Formula fed at 6 months	-	-	.09 [†]	.05	-	-	-.13 [†]	.08
Breastfed at 12 months	-	-	-.07	.06	-	-	.17	.10
Formula fed at 12 months	-	-	-.04	.05	-	-	.11	.09
Other milk fed at 12 months	-	-	-.06	.05	-	-	.05	.09
Healthy foods consumed per day	-	-	.01	.03	-	-	-.01	.05
Unhealthy foods consumed per day	-	-	.00	.03	-	-	-.04	.06
Infant sex	-	-	.00	.03	-	-	.01	.05
Infant race	-	-	.01	.03	-	-	-.03	.05
Infant ethnicity	-	-	.07	.03	-	-	-.08	.06
Number of epochs	-	-	.10*	.03	-	-	-.21**	.05
Cohort	-	-	.00	.04	-	-	.03	.07
Adjusted R ² = 0.79					Adjusted R ² = 0.51			

Note. Breastfed (1 =exclusively breastmilk fed, 0 =mixed breast milk, formula or other-milk fed); formula-fed (1 =exclusively formula fed, 0 =exclusively breastmilk fed, 0 =mixed breast milk, formula or other-milk fed); Other milk fed (1 =exclusively other milk fed, 0 =exclusively breastmilk fed, 0 =exclusively formula fed, 0 =mixed breast milk, formula or other-milk fed); Infant sex (0 =male, 1 =female); Infant race (0 =White, 1 =Non-white); Infant ethnicity (0 =Non-Hispanic/Latino, 1 =Hispanic/Latino); Cohort (0 =Cohort 1, 1 =Cohort 2); ** $p < .01$ * $p < 0.05$ [†] $p < .10$.

Huttenlocher and Dabholkar, 1997). Future work employing both EEG and MRI will be helpful in elucidating the neural mechanisms underlying changes in EEG power during infancy.

Another important consideration is that the observed age-related patterns in relative power may be driven, in part, by changes in the aperiodic components of the EEG signal (Ostlund et al., 2022). EEG power reflects a combination of aperiodic activity (a 1/f-like background spectrum) and periodic or oscillatory activity (peaks superimposed on this background; Donoghue et al., 2020). Recent work has shown that periodic and relative power trajectories may diverge, and that developmental changes in relative theta and beta power may be more likely to be influenced by age-related variation in the aperiodic signal (e.g., Rico-Picó et al., 2023). As such, the age-related patterns observed in our study could be driven by periodic activity, aperiodic activity, or both. Future studies using spectral parameterization methods (e.g., Donoghue et al., 2020; Hill et al., 2022; Wilkinson et al., 2024) will be important for clarifying the underlying components that contribute to developmental changes in EEG power during infancy.

Findings from the current study also align with previous research linking postnatal family SES to developmental differences in lower- and higher-frequency brain activity in infants (Brito et al., 2020; Otero, 1997; Otero et al., 2003; Pierce et al., 2019), and extends this work by showing that *prenatal* family income may shape age-related patterns of brain activity during the first 18 months of life. Although resting EEG power in infants is partially heritable (Orekhova et al., 2003), it is also subject to substantial environmental influence (Saby and Marshall, 2012; Troller-Renfree et al., 2020). The patterns of brain activity we observed may therefore reflect genetic predispositions and/or more proximal prenatal or early postnatal experiences that covary with SES (Hackman et al., 2010). Our sensitivity analyses, however, indicated that associations between prenatal family ITN and age-related patterns of brain activity were independent of various measured prenatal factors (maternal stress, maternal vocabulary), postnatal factors (maternal stress, feeding style, engagement in learning activities, home language input, nutrition), and infant characteristics (demographic factors, gestational age, birth weight). These associations were also specific to prenatal, and not postnatal ITN. Nonetheless, because we did not directly assess genetic factors or measure other important prenatal (e.g., maternal nutrition, exercise, sleep quality, toxin exposure) and postnatal (e.g., parenting, childcare) experiences, their potential influence on the development of infant brain activity cannot be ruled out.

The patterns of brain activity observed among infants from lower-income families may have important implications for their future developmental outcomes. In infancy, both a greater amount of lower-frequency (theta) power, as well as less higher-frequency (beta) power, have been linked to lower scores on subsequent cognitive (Scher et al., 1996; Williams et al., 2012) and socio-emotional skills (Brito et al., 2019), as well as with the development of psychopathology (McLaughlin et al., 2010). Such patterns of brain activity in infancy also mediate associations between environmental deprivation and later outcomes, such as lower IQ (Tan et al., 2023) and ADHD symptoms (McLaughlin et al., 2010), suggesting a potential pathway linking prenatal income disparities to increased risk for adverse developmental and mental health outcomes. We note here, however, that children's brain development reflects an adaptation to their lived experiences (Noble et al., 2021), and not all children from families facing socioeconomic disadvantage show the brain activity patterns we observed. Moreover, such patterns of brain activity may support the development of skills that are advantageous in the context of limited material resources (Ellis et al., 2022). Thus, adaptation does not necessarily indicate dysfunction or deficit but rather an expected and appropriate response that helps children function within their environment.

Interestingly, prenatal family income—but not parental education—was associated with age-related patterns of resting brain activity in infants. These data are consistent with the results of some previous studies (Cantiani et al., 2019; Tomalski et al., 2013; Wilkinson et al.,

2023) and underscore the fact that family income and parental education represent distinct aspects of SES, each tied to different early experiences (Duncan and Magnuson, 2012). It is possible that experiences associated with income (e.g., crowding, neighborhood poverty, exposure to stress and toxins, nutrition, access to medical care) may have a stronger impact on age-related patterns of brain activity in early childhood (Troller-Renfree et al., 2022). In contrast, parental education might more directly shape the parenting environment, potentially influencing these patterns later in childhood.

Our follow-up exploratory analyses indicated some regional specificity. For instance, higher prenatal family ITN was associated with a more rapid decrease in lower-frequency (relative theta) power over time, particularly in the frontal, temporal, and occipital regions; however, these associations were not statistically significant after adjusting for multiple comparisons. In contrast, higher prenatal family ITN was linked to a more rapid increase in higher-frequency (relative beta) power across multiple regions, including the frontal, central, parietal, temporal, and occipital areas. These results align with prior research indicating socioeconomic-related differences in regional theta (Otero, 1997; Otero et al., 2003; Sandre et al., 2024) and beta (Brito et al., 2020; Jensen et al., 2021; Otero, 1994) power in young children. Taken together, these findings suggest that family income may influence functional brain development in a region-specific manner, with certain regions showing greater sensitivity to income-related differences in specific frequency bands.

To our knowledge, this is one of the first longitudinal studies to examine both prenatal family ITN and parental education and their associations with age-related patterns of resting brain activity in a large, socioeconomically and racially/ethnically diverse sample of infants. However, there are important limitations to consider. First, this study is correlational, and we cannot establish that observed differences in infant brain activity are caused by prenatal family SES. Second, the effect sizes observed in this study are small in magnitude, highlighting that prenatal family income is likely only one of many factors associated with differences in the development of infant brain activity. Third, due to the COVID-19 pandemic, EEG data could not be collected from the full sample at the 6-, 12-, and 15/18-month assessments, limiting statistical power and the generalizability of our findings. We therefore conducted sensitivity analyses restricted to Cohort 2, as this subset includes participants for whom EEG data collection resumed post-pandemic and comprises a higher proportion of infants with repeated measures. Results from this analysis were consistent with the full-sample findings (see [Supplementary Table S10](#) and [Fig. S18](#)). Although the interaction between age and prenatal ITN on relative theta power was marginally significant in this subset, the overall pattern of associations supports the robustness of our main findings. Fourth, we observed differences between families recruited before and after the pandemic on several study variables (e.g., maternal prenatal perceived stress, physiological stress, receptive vocabulary scores, parental education, infant ethnicity, and age), making it unclear whether these differences influenced the observed results. Although we controlled for these variables in our sensitivity analyses, future studies are needed to determine whether our results replicate in samples with more complete data and to better understand the potential influence of the pandemic on age-related patterns of resting brain activity in infancy. Finally, the frequency bands used in this study were selected based on prior work in infants (Tomalski et al., 2013) and to maintain consistency across analyses. However, we acknowledge that definitions of frequency bands—particularly in the upper beta and gamma ranges—vary considerably across the infant resting EEG literature (e.g., Anderson and Perone, 2018; Troller-Renfree et al., 2025). These definitional inconsistencies may contribute to differences in findings across studies and should be considered when interpreting our results.

In conclusion, this study suggests that resting brain activity follows non-linear age-related patterns during the first 18 months of life. Moreover, prenatal ITN is associated with differences in these patterns,

suggesting that prenatal environments may shape early functional brain development. Notably, SES-related differences in patterns of infant brain activity are specific to family income, rather than parental education, and are independent of various measured prenatal and postnatal experiences, as well as infant demographic and health-related factors. While additional research is needed, our results suggest the possibility that investment in income-support programs and policies during pregnancy may be an important target for supporting infants' brain development.

CRedit authorship contribution statement

Aislinn Sandre: Writing – original draft, Visualization, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation. **Sonya V. Troller-Renfree:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation. **Melina Amarante:** Writing – review & editing, Investigation, Formal analysis. **Amanda M. Dettmer:** Writing – review & editing, Formal analysis. **Jerrold S. Meyer:** Writing – review & editing, Formal analysis. **Kimberly G. Noble:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank all of our research participants and their families for their time and participation. Finally, thanks to Olivia Colon, Isabel Kovacs, Dayanara Sanchez and Montse Lopez Orozco for their help with data collection.

Funding

This work was supported by the Eunice Kennedy Shriver National Institute of Child Health and Human Development R01HD093707 awarded to Kimberly Noble and K99HD104923 awarded to Sonya Troller-Renfree, and the Canadian Institutes of Health Research Fellowship (187925) awarded to Aislinn Sandre. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Data Statement

The study's deidentified data as well as syntax used for all analyses are available at the following link: <https://osf.io/md7fh/>. The code and stimuli for the resting EEG tasks are available from the corresponding author upon request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.dcn.2025.101611](https://doi.org/10.1016/j.dcn.2025.101611).

Data availability

The study's deidentified data as well as syntax used for all analyses are available at the following link: <https://osf.io/md7fh/>

References

- Anderson, A.J., Perone, S., 2018. Developmental change in the resting state electroencephalogram: insights into cognition and the brain. *Brain Cogn.* 126, 40–52. <https://doi.org/10.1016/j.bandc.2018.08.001>.
- Arnett, A.B., Guiney, H., Bakir-Demir, T., Trudgen, A., Schierding, W., Reid, V., O'Sullivan, J., Gluckman, P., Reese, E., Poulton, R., 2024. Resting EEG correlates of neurodevelopment in a socioeconomically and linguistically diverse sample of toddlers: wave 1 of the kia Timata pai best start New Zealand study. *Dev. Cogn. Neurosci.* 65, 101336. <https://doi.org/10.1016/j.dcn.2023.101336>.
- Benninger, C., Matthis, P., Scheffner, D., 1984. EEG development of healthy boys and girls. Results of a longitudinal study. *Electroencephalogr. Clin. Neurophysiol.* 57 (1), 1–12.
- Bradley, R.H., Corwyn, R.F., 2002. Socioeconomic status and child development. *Annu. Rev. Psychol.* 53 (1), 371–399. <https://doi.org/10.1146/annurev.psych.53.100901.135233>.
- Brito, N.H., Elliott, A.J., Isler, J.R., Rodriguez, C., Friedrich, C., Shuffrey, L.C., Fifer, W. P., 2019. Neonatal EEG linked to individual differences in socioemotional outcomes and autism risk in toddlers. *Dev. Psychobiol.* 61 (8), 1110–1119. <https://doi.org/10.1002/dev.21870>.
- Brito, N.H., Fifer, W.P., Myers, M.M., Elliott, A.J., Noble, K.G., 2016. Associations among family socioeconomic status, EEG power at birth, and cognitive skills during infancy. *Dev. Cogn. Neurosci.* 19, 144–151. <https://doi.org/10.1016/j.dcn.2016.03.004>.
- Brito, N.H., Troller-Renfree, S.V., Leon-Santos, A., Isler, J.R., Fifer, W.P., Noble, K.G., 2020. Associations among the home language environment and neural activity during infancy. *Dev. Cogn. Neurosci.* 43, 100780. <https://doi.org/10.1016/j.dcn.2020.100780>.
- Cantiani, C., Piazza, C., Mornati, G., Molteni, M., Riva, V., 2019. Oscillatory gamma activity mediates the pathway from socioeconomic status to language acquisition in infancy. *Infant Behav. Dev.* 57, 101384. <https://doi.org/10.1016/j.infbeh.2019.101384>.
- Chen, H., Liu, J., Chen, Y., Salzwedel, A., Cornea, E., Gilmore, J.H., Gao, W., 2021. Developmental heatmaps of brain functional connectivity from newborns to 6-year-olds. *Dev. Cogn. Neurosci.* 50, 100976. <https://doi.org/10.1016/j.dcn.2021.100976>.
- Clarke, A.R., Barry, R.J., McCarthy, R., Selikowitz, M., 2001. Age and sex effects in the EEG: development of the normal child. *Clin. Neurophysiol.* 112 (5), 806–814. [https://doi.org/10.1016/S1388-2457\(01\)00488-6](https://doi.org/10.1016/S1388-2457(01)00488-6).
- Cohen, S., Kamarck, T., Mermelstein, R., 1983. A global measure of perceived stress. *J. Health Soc. Behav.* 24 (4), 385–396. <https://doi.org/10.2307/2136404>.
- Cuevas, K., Bell, M.A., 2022. EEG frequency development across infancy and childhood. In: Gable, In.P., Miller, M., Bernat, E. (Eds.), *Oxford Handbook of Human EEG Frequency Analysis*. Oxford University Press, pp. 293–323.
- Cuffin, B.N., 1993. Effects of local variations in skull and scalp thickness on EEG's and MEG's. *IEEE Trans. Biomed. Eng.* 40 (1), 42–48. <https://doi.org/10.1109/10.204770>.
- Davis Research. (2017). Los Angeles County WIC survey. Retrieved from: (<https://lawic.data.org/wp-content/uploads/2014/09/WIC-Parents-Quex-English-FINAL.pdf>).
- Dawson, G., Ashman, S.B., Carver, L.J., 2000. The role of early experience in shaping behavioral and brain development and its implications for social policy. *Dev. Psychopathol.* 12 (4), 695–712. <https://doi.org/10.1017/S0954579400004089>.
- Debnath, R., Buzzell, G.A., Morales, S., Bowers, M.E., Leach, S.C., Fox, N.A., 2020. The maryland analysis of developmental EEG (MADE) pipeline. *Psychophysiology* 57 (6), e13580. <https://doi.org/10.1111/psyp.13580>.
- Delorme, A., Makeig, S., 2004. EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *J. Neurosci. Methods* 134 (1), 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>.
- Donoghue, T., Haller, M., Peterson, E.J., Varma, P., Sebastian, P., Gao, R., Noto, T., Lara, A.H., Wallis, J.D., Knight, R.T., 2020. Parameterizing neural power spectra into periodic and aperiodic components. *Nat. Neurosci.* 23 (12), 1655–1665. <https://doi.org/10.1038/s41593-020-00744-x>.
- Dubois, J., Dehaene-Lambertz, G., Kulikova, S., Poupon, C., Hüppi, P.S., Hertz-Pannier, L., 2014. The early development of brain white matter: a review of imaging studies in fetuses, newborns and infants. *Neuroscience* 276, 48–71. <https://doi.org/10.1016/j.neuroscience.2013.12.044>.
- Duncan, G.J., Magnuson, K., 2012. Socioeconomic status and cognitive functioning: moving from correlation to causation. *Wiley Interdiscip. Rev. Cognit. Sci.* 3 (3), 377–386. <https://doi.org/10.1002/wcs.1176>.
- Ellis, B.J., Abrams, L.S., Masten, A.S., Sternberg, R.J., Tottenham, N., Frankenhuys, W.E., 2022. Hidden talents in harsh environments. *Dev. Psychopathol.* 34 (1), 95–113. <https://doi.org/10.1017/S0954579420000887>.
- Farah, M.J., 2017. The neuroscience of socioeconomic status: correlates, causes, and consequences. *Neuron* 96 (1), 56–71. <https://doi.org/10.1016/j.neuron.2017.08.034>.
- Gershon, R.C., Cook, K.F., Mungas, D., Manly, J.J., Slotkin, J., Beaumont, J.L., Weintraub, S., 2014. Language measures of the NIH toolbox cognition battery. *J. Int. Neuropsychol. Soc.* 20 (6), 642–651. <https://doi.org/10.1017/S1355617714000411>.
- Gilmore, J.H., Knickmeyer, R.C., Gao, W., 2018. Imaging structural and functional brain development in early childhood. *Nat. Rev. Neurosci.* 19 (3), 123–137. <https://doi.org/10.1038/nrn.2018.1>.
- Gou, Z., Choudhury, N., Benasich, A.A., 2011. Resting frontal gamma power at 16, 24 and 36 months predicts individual differences in language and cognition at 4 and 5 years. *Behav. Brain Res.* 220 (2), 263–270. <https://doi.org/10.1016/j.bbr.2011.01.048>.
- Greff, M.J.E., Levine, J.M., Abuzgaia, A.M., Elzagallaai, A.A., Rieder, M.J., van Uum, S.H. M., 2019. Hair cortisol analysis: an update on methodological considerations and

- clinical applications. *Clin. Biochem.* 63, 1–9. <https://doi.org/10.1016/j.clinbiochem.2018.09.010>.
- Hackman, D.A., Farah, M.J., Meaney, M.J., 2010. Socioeconomic status and the brain: mechanistic insights from human and animal research. *Nat. Rev. Neurosci.* 11 (9), 651–659. <https://doi.org/10.1038/nrn2897>.
- Hagemann, D., Hewig, J., Walter, C., Naumann, E., 2008. Skull thickness and magnitude of EEG alpha activity. *Clin. Neurophysiol.* 119 (6), 1271–1280. <https://doi.org/10.1016/j.clinph.2008.02.010>.
- Harmony, T., Alvarez, A., Pascual, R., Ramos, A., Marosi, E., Díaz De León, A.E., Valdés, P., Becker, J., 1988. EEG maturation on children with different economic and psychosocial characteristics. *Int. J. Neurosci.* 41 (1–2), 103–113. <https://doi.org/10.3109/00207458808985747>.
- Herzberg, M.P., Smyser, C.D., 2024. Prenatal social determinants of health: narrative review of maternal environments and neonatal brain development. *Pediatr. Res.* 96 (6), 1417–1428. <https://doi.org/10.1038/s41390-024-03345-7>.
- Hill, A.T., Clark, G.M., Bigelow, F.J., Lum, J.A.G., Enticott, P.G., 2022. Periodic and aperiodic neural activity displays age-dependent changes across early-to-middle childhood. *Dev. Cogn. Neurosci.* 54, 101076. <https://doi.org/10.1016/j.dcn.2022.101076>.
- Hoff, E., Laursen, B., 2019. Socioeconomic status and parenting. In: Bornstein, In.M.H. (Ed.), *Handbook of Parenting: Biology and ecology of parenting* (3rd ed. Routledge/Taylor & Francis Group, pp. 421–447. <https://doi.org/10.4324/9780429401459-13>.
- Hofstee, M., Endendijk, J., Huijding, J., van der Velde, B., Vidal, J., Deković, M., 2024. Maturation changes in frontal EEG alpha and theta activity from infancy into early childhood and the relation with self-regulation in boys and girls. *Dev. Cogn. Neurosci.* 70, 101445. <https://doi.org/10.1016/j.dcn.2024.101445>.
- Hu, D.K., Goetz, P.W., To, P.D., Garner, C., Magers, A.L., Skora, C., Tran, N., Yuen, T., Hussain, S.A., Shrey, D.W., 2022. Evolution of cortical functional networks in healthy infants. *Front. Netw. Physiol.* 2, 893826. <https://doi.org/10.3389/fnetp.2022.893826>.
- Huttenlocher, P.R., Dabholkar, A.S., 1997. Regional differences in synaptogenesis in human cerebral cortex. *J. Comp. Neurol.* 387 (2), 167–178. [https://doi.org/10.1002/\(SICI\)1096-9861\(19971020\)387:2<167::AID-CNEI>3.0.CO;2-Z](https://doi.org/10.1002/(SICI)1096-9861(19971020)387:2<167::AID-CNEI>3.0.CO;2-Z).
- Jensen, S.K.G., Xie, W., Kumar, S., Haque, R., Petri, W.A., Nelson, I.I., C.A., 2021. Associations of socioeconomic and other environmental factors with early brain development in Bangladeshi infants and children. *Dev. Cogn. Neurosci.* 50, 100981. <https://doi.org/10.1016/j.dcn.2021.100981>.
- Kim, J.H., 2019. Multicollinearity and misleading statistical results. *Korean J. Anesthesiol.* 72 (6), 558–569. <https://doi.org/10.4097/kja.19087>.
- Leach, S.C., Morales, S., Bowers, M.E., Buzzell, G.A., Debnath, R., Beall, D., Fox, N.A., 2020. Adjusting ADJUST: optimizing the ADJUST algorithm for pediatric data using geodesic nets. *Psychophysiology* 57 (8), e13566. <https://doi.org/10.1111/psyp.13566>.
- Little, R.J.A., 1988. A test of missing completely at random for multivariate data with missing values. *J. Am. Stat. Assoc.* 83, 1198. <https://doi.org/10.2307/2290157>.
- Liu, J., Chen, H., Cornea, E., Gilmore, J.H., Gao, W., 2023. Longitudinal developmental trajectories of functional connectivity reveal regional distribution of distinct age effects in infancy. *Cereb. Cortex* 33 (19), 10367–10379. <https://doi.org/10.1093/cercor/bhad288>.
- Maguire, M.J., Schneider, J.M., 2019. Socioeconomic status related differences in resting state EEG activity correspond to differences in vocabulary and working memory in grade school. *Brain Cogn.* 137, 103619. <https://doi.org/10.1016/j.bandc.2019.103619>.
- Marshall, P.J., Bar-Haim, Y., Fox, N.A., 2002. Development of the EEG from 5 months to 4 years of age. *Clin. Neurophysiol.* 113 (8), 1199–1208. [https://doi.org/10.1016/S1388-2457\(02\)00163-3](https://doi.org/10.1016/S1388-2457(02)00163-3).
- Marshall, P.J., Fox, N.A., Group, B.C., 2004. A comparison of the electroencephalogram between institutionalized and community children in Romania. *J. Cogn. Neurosci.* 16 (8), 1327–1338. <https://doi.org/10.1162/089929042304723>.
- Marshall, P.J., Reeb, B.C., Fox, N.A., Nelson, C.A., Zeanah, C.H., 2008. Effects of early intervention on EEG power and coherence in previously institutionalized children in Romania. *Dev. Psychopathol.* 20 (3), 861–880. <https://doi.org/10.1017/S0954579408000412>.
- McLaughlin, K.A., Fox, N.A., Zeanah, C.H., Sheridan, M.A., Marshall, P., Nelson, C.A., 2010. Delayed maturation in brain electrical activity partially explains the association between early environmental deprivation and symptoms of attention-deficit/hyperactivity disorder. *Biol. Psychiatry* 68 (4), 329–336. <https://doi.org/10.1016/j.biopsych.2010.04.005>.
- Merz, E.C., Desai, P.M., Maskus, E.A., Melvin, S.A., Rehman, R., Torres, S.D., Meyer, J., He, X., Noble, K.G., 2019. Socioeconomic disparities in chronic physiologic stress are associated with brain structure in children. *Biol. Psychiatry* 86 (12), 921–929. <https://doi.org/10.1016/j.biopsych.2019.05.024>.
- Monk, C., Lugo-Candelas, C., Trumpff, C., 2019. Prenatal developmental origins of future psychopathology: mechanisms and pathways. *Annu. Rev. Clin. Psychol.* 15 (1), 317–344. <https://doi.org/10.1146/annurev-clinpsy-050718-095539>.
- Nielsen, A.N., Triplett, R.L., Bernardez, L.M., Tooley, U.A., Herzberg, M.P., Lean, R.E., Kaplan, S., Meyer, D., Kenley, J.K., Alexopoulos, D., 2024. Prenatal social disadvantage is associated with alterations in functional networks at birth. *Proc. Natl. Acad. Sci. USA* 121 (50), e2405448121. <https://doi.org/10.1073/pnas.2405448121>.
- Noble, K.G., Hart, E.R., Sperber, J.F., 2021. Socioeconomic disparities and neuroplasticity: moving toward adaptation, intersectionality, and inclusion. *Am. Psychol.* 76 (9), 1486. <https://doi.org/10.1037/amp0000934>.
- Nolan, H., Whelan, R., Reilly, R.B., 2010. FASTER: fully automated statistical thresholding for EEG artifact rejection. *J. Neurosci. Methods* 192 (1), 152–162. <https://doi.org/10.1016/j.jneumeth.2010.07.015>.
- Orehkova, E.V., Stroganova, T.A., Posikera, I.N., Malykh, S.B., 2003. Heritability and “environmentability” of electroencephalogram in infants: the twin study. *Psychophysiology* 40 (5), 727–741. <https://doi.org/10.1111/1469-8986.00073>.
- Ostlund, B., Donoghue, T., Anaya, B., Gunther, K.E., Karalunas, S.L., Voytek, B., Pérez-Edgar, K.E., 2022. Spectral parameterization for studying neurodevelopment: how and why. *Dev. Cogn. Neurosci.* 54, 101073. <https://doi.org/10.1016/j.dcn.2022.101073>.
- Otero, G.A., 1994. EEG spectral analysis in children with sociocultural handicaps. *Int. J. Neurosci.* 79 (3–4), 213–220. <https://doi.org/10.3109/00207459408986082>.
- Otero, G.A., 1997. Poverty, cultural disadvantage and brain development: a study of pre-school children in Mexico. *Electroencephalogr. Clin. Neurophysiol.* 102 (6), 512–516. [https://doi.org/10.1016/s0013-4694\(97\)95213-9](https://doi.org/10.1016/s0013-4694(97)95213-9).
- Otero, G.A., Pliego-Rivero, F.B., Fernández, T., Ricardo, J., 2003. EEG development in children with sociocultural disadvantages: a follow-up study. *Clin. Neurophysiol.* 114 (10), 1918–1925. [https://doi.org/10.1016/S1388-2457\(03\)00173-1](https://doi.org/10.1016/S1388-2457(03)00173-1).
- Perone, S., Palanisamy, J., Carlson, S.M., 2018. Age-related change in brain rhythms from early to middle childhood: links to executive function. *Dev. Sci.* 21 (6), e12691. <https://doi.org/10.1111/desc.12691>.
- Pierce, L.J., Thompson, B.L., Gharib, A., Schlueter, L., Reilly, E., Valdes, V., Roberts, S., Conroy, K., Levitt, P., Nelson, C.A., 2019. Association of perceived maternal stress during the perinatal period with electroencephalography patterns in 2-month-old infants. *JAMA Pediatr.* 173 (6), 561–570. <https://doi.org/10.1001/jamapediatrics.2019.0492>.
- Rico-Picó, J., Moyano, S., Conejero, Á., Hoyo, Á., Ballesteros-Duperón, M.Á., Rueda, M. R., 2023. Early development of electrophysiological activity: contribution of periodic and aperiodic components of the EEG signal. *Psychophysiology* 60 (11), e14360. <https://doi.org/10.1111/psyp.14360>.
- Rodriguez, E.T., Tamis-LeMonda, C.S., 2011. Trajectories of the home learning environment across the first 5 years: associations with children’s vocabulary and literacy skills at prekindergarten. *Child Dev.* 82 (4), 1058–1075. <https://doi.org/10.1111/j.1467-8624.2011.01614.x>.
- Saby, J.N., Marshall, P.J., 2012. The utility of EEG band power analysis in the study of infancy and early childhood. *Dev. Neuropsychol.* 37 (3), 253–273. <https://doi.org/10.1080/87565641.2011.614663>.
- Sandre, A., Troller-Renfree, S.V., Giebler, M.A., Meyer, J.S., Noble, K.G., 2024. Prenatal family income, but not parental education, is associated with resting brain activity in 1-month-old infants. *Sci. Rep.* 14 (1), 13638. <https://doi.org/10.1038/s41598-024-64498-3>.
- Sarason, I.G., Johnson, J.H., Siegel, J.M., 1978. Assessing the impact of life changes: development of the life experiences survey. *J. Consult. Clin. Psychol.* 46 (5), 932–946. <https://doi.org/10.1037/0022-006x.46.5.932>.
- Scher, M.S., Steppe, D.A., Banks, D.L., 1996. Prediction of lower developmental performances of healthy neonates by neonatal EEG-sleep measures. *Pediatr. Neurol.* 14 (2), 137–144. [https://doi.org/10.1016/0887-8994\(96\)00013-6](https://doi.org/10.1016/0887-8994(96)00013-6).
- Simon, K.R., Merz, E.C., He, X., Desai, P.M., Meyer, J.S., Noble, K.G., 2021. Socioeconomic factors, stress, hair cortisol, and White matter microstructure in children. *Dev. Psychobiol.* 63 (6), e22147. <https://doi.org/10.1002/dev.22147>.
- Simpson, G.L., 2024. Gratia: an r package for exploring generalized additive models. *J. Open Source Softw.* 9 (104), 6962. <https://doi.org/10.21105/joss.06962>.
- Sperber, J.F., Hart, E.R., Troller-Renfree, S.V., Watts, T.W., Noble, K.G., 2023. The effect of the COVID-19 pandemic on infant development and maternal mental health in the first 2 years of life. *Infancy* 28 (1), 107–135. <https://doi.org/10.1111/inf.12511>.
- Takano, T., Ogawa, T., 1998. Characterization of developmental changes in EEG-gamma band activity during childhood using the autoregressive model. *Pediatr. Int.* 40 (5), 446–452. <https://doi.org/10.1111/j.1442-200X.1998.tb01966.x>.
- Tan, E., Tang, A., Debnath, R., Humphreys, K.L., Zeanah, C.H., Nelson, C.A., Fox, N.A., 2023. Resting brain activity in early childhood predicts IQ at 18 years. *Dev. Cogn. Neurosci.* 63, 101287. <https://doi.org/10.1016/j.dcn.2023.101287>.
- Tan, E., Troller-Renfree, S.V., Morales, S., Buzzell, G.A., McSweeney, M., Antúnez, M., Fox, N.A., 2024. Theta activity and cognitive functioning: integrating evidence from resting-state and task-related developmental electroencephalography (EEG) research. *Dev. Cogn. Neurosci.* 67, 101404. <https://doi.org/10.1016/j.dcn.2024.101404>.
- Tierney, A.L., Gabard-Durnam, L., Vogel-Farley, V., Tager-Flusberg, H., Nelson, C.A., 2012. Developmental trajectories of resting EEG power: an endophenotype of autism spectrum disorder. *PLOS One* 7 (6), e39127. <https://doi.org/10.1371/journal.pone.0039127>.
- Tomalski, P., Moore, D.G., Ribeiro, H., Axelsson, E.L., Murphy, E., Karmiloff-Smith, A., Johnson, M.H., Kushnerenko, E., 2013. Socioeconomic status and functional brain development—associations in early infancy. *Dev. Sci.* 16 (5), 676–687. <https://doi.org/10.1111/desc.12079>.
- Troller-Renfree, S.V., Costanzo, M.A., Duncan, G.J., Magnuson, K., Gennetian, L.A., Yoshikawa, H., Halpern-Meekin, S., Fox, N.A., Noble, K.G., 2022. The impact of a poverty reduction intervention on infant brain activity. *Proc. Natl. Acad. Sci. USA* 119 (5), e2115649119. <https://doi.org/10.1073/pnas.2115649119>.
- Troller-Renfree, S.V., Gray, K.N., Sandre, A., Hart, E.R., Sperber, J.F., Dettmer, A.M., Meyer, J.S., Noble, K.G., 2025. Associations between prenatal maternal stress and infant resting brain activity: a preregistered investigation. *Dev. Psychobiol.*
- Troller-Renfree, S.V., Sperber, J.F., Hart, E.R., Costanzo, M.A., Gennetian, L.A., Meyer, J. S., Fox, N.A., Noble, K.G., 2023. Associations between maternal stress and infant resting brain activity among families residing in poverty in the US. *Biol. Psychol.* 184, 108683. <https://doi.org/10.1016/j.biopsycho.2023.108683>.
- Troller-Renfree, S.V., Brito, N.H., Desai, P.M., Leon-Santos, A.G., Wiltshire, C.A., Motton, S.N., Meyer, J.S., Isler, J., Fifer, W.P., Noble, K.G., 2020. Infants of mothers with higher physiological stress show alterations in brain function. *Dev. Sci.* 23 (6), e12976. <https://doi.org/10.1111/desc.12976>.

- Troller-Renfree, S.V., Morales, S., Buzzell, G.A., Sandre, A., 2025. Heterogeneity in pediatric resting EEG data processing and analysis: a state of the field. *Psychophysiology* 62 (2), e14733. <https://doi.org/10.1111/psyp.14733>.
- Troller-Renfree, S.V., Morales, S., Leach, S.C., Bowers, M.E., Debnath, R., Fifer, W.P., Fox, N.A., Noble, K.G., 2021. Feasibility of assessing brain activity using mobile, in-home collection of electroencephalography: methods and analysis. *Dev. Psychobiol.* 63 (6), e22128. <https://doi.org/10.1002/dev.22128>.
- Ursache, A., Merz, E.C., Melvin, S., Meyer, J., Noble, K.G., 2017. Socioeconomic status, hair cortisol and internalizing symptoms in parents and children. *Psychoneuroendocrinology* 78, 142–150. <https://doi.org/10.1016/j.psycheneu.2017.01.020>.
- Van den Bergh, B.R.H., van den Heuvel, M.I., Lahti, M., Braeken, M., de Rooij, S.R., Entringer, S., Hoyer, D., Roseboom, T., Räikkönen, K., King, S., 2020. Prenatal developmental origins of behavior and mental health: the influence of maternal stress in pregnancy. *Neurosci. Biobehav. Rev.* 117, 26–64. <https://doi.org/10.1016/j.neubiorev.2017.07.003>.
- Von Stein, A., Sarnthein, J., 2000. Different frequencies for different scales of cortical integration: from local gamma to long range alpha/theta synchronization. *Int. J. Psychophysiol.* 38 (3), 301–313. [https://doi.org/10.1016/S0167-8760\(00\)00172-0](https://doi.org/10.1016/S0167-8760(00)00172-0).
- Wilkinson, C.L., Pierce, L., Sideridis, G., Wade, M., Nelson, C.A., 2023. Associations between EEG trajectories, family income, and cognitive abilities over the first two years of life. *Dev. Cogn. Neurosci.* 61, 101260. <https://doi.org/10.1016/j.dcn.2023.101260>.
- Wilkinson, C.L., Yankowitz, L.D., Chao, J.Y., Gutiérrez, R., Rhoades, J.L., Shinnar, S., Purdon, P.L., Nelson, C.A., 2024. Developmental trajectories of EEG aperiodic and periodic components in children 2–44 months of age. *Nat. Commun.* 15 (1), 5788. <https://doi.org/10.1038/s41467-024-50204-4>.
- Williams, I.A., Tarullo, A.R., Grieve, P.G., Wilpers, A., Vignola, E.F., Myers, M.M., Fifer, W.P., 2012. Fetal cerebrovascular resistance and neonatal EEG predict 18-month neurodevelopmental outcome in infants with congenital heart disease. *Ultrasound Obstet. Gynecol.* 40 (3), 304–309. <https://doi.org/10.1002/uog.11144>.
- Wood, S.N., 2004. Stable and efficient multiple smoothing parameter estimation for generalized additive models. *J. Am. Stat. Assoc.* 99 (467), 673–686. <https://doi.org/10.1198/016214504000000980>.
- Wood, S.N. 2017. Generalized additive models: an introduction with R. Chapman and Hall/CRC.